B.TECH ARTIFICIAL INTELLIGENCE & MACHINE LEARNING · CA2 CAPSTONE PROJECT

Decision Debt OS

Turning business decisions into measurable learning



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Division: [Division] **Institution:** [Institution Name] **Academic Year:** 2026–27

THE PROBLEM

Decisions get made. Assumptions get forgotten.



Dashboards and BI tools show what happened. Project trackers show what is scheduled. Neither one records **what a team believed would happen** before they acted — so when the belief turns out to be wrong, nothing captures that it was ever tested, and the same unexamined assumption drives the next decision.

CORE INSIGHT

A decision is not finished when it is made.

> *“A business decision is incomplete until its original assumption is compared with the real-world outcome.”*

This gap has a name:

Decision Debt — the accumulation of decisions whose assumptions were never checked against reality.

DEFINING THE PROBLEM

What Is Decision Debt?

Unresolved Assumptions

A decision was made, but its underlying belief was never written down in a checkable form.

Overdue Reviews

A deadline for checking the outcome has passed, and no one has looked at the result.

Invalidated Assumptions Still Driving Work

The assumption was checked, found false, but the decision or policy it justified is still active.

Repeated Forecasting Errors

The same type of assumption keeps failing across different decisions, and no one notices the pattern.

THE SOLUTION

Capture → Structure → Forecast → Measure → Evaluate → Learn



UNSTRUCTURED INPUT

"Let's offer the 10% discount to high-risk customers — I think it'll cut churn to around 18% within a month."

STRUCTURED DECISION RECORD

INTERVENTION	10% retention discount
TARGET METRIC	30-day churn, high-risk cohort
FORECAST	25% → 18%
CONFIDENCE	62%
DEADLINE	+30 days